

GTÜ CSE 495&496

P-Wave Detector For Earthquakes

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GEBZE
TEKNİK ÜNİVERSİTESİ



Content

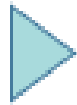
- Scheme and Description of the Project
- Timeline
- Success Criteria

Scheme and Description of the Project

- In earthquake waveforms, P-wave, S-wave and surface wave are the three main groups of wave trains. P-wave is the first wave group to arrive in an earthquake due to its faster propagation speed than S-wave and surface wave.
- So are trying to implement a machine learning model for p-wave detection for the early detection of earthquakes.

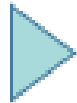
Time-line

05.04.2023



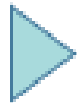
- Searching about p-wave
- Searching dataset
- Look for similar projects

First Representation



- Learning machine learn algorithms
- Creating a model and test

Mid Representation



- Test
- Result
- Improving

Final Presentation

Success Criteria

- We want to detect the earthquakes 3 seconds ago before it starts
- 80% accuracy rate for true detection
- Use at least 5 years data (

<https://www.kaggle.com/datasets/omercolakoglu/turkey-earthquake-data-for-20-years>

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